

The Impact of COVID-19 on the Incidence of Lower Extremity Amputations in Diabetic Patients

Abstract

The purpose of this study is to investigate the impact of the COVID-19 pandemic on the incidence of lower body amputations in patients diagnosed with diabetes mellitus. Specifically, we explored the variability of amputation rates across different hospitals and counties in the state of New York before and after the outbreak of the COVID-19 pandemic.

Lower extremity amputations (LEAs) are a common surgical procedure used to treat diabetic foot ulcers, a severe complication of diabetes mellitus. LEAs, whether minor or major, involve the removal of diseased tissue to retain the extremity's length [22]. Both minor and major amputations are severe interventions, with 5-year mortality rates ranging from 29% to 69% for minor amputations and 52% to 80% for major amputations, respectively [23]. Previous studies have examined amputation rates over time to investigate the potential altering effects of the COVID-19 pandemic on chronic care delivery [24-27]. Amputation rates among diabetic patients vary across hospitals and counties.

This study focuses on data from hospitals in New York State before and after 2020, the first year of the COVID-19 pandemic. Furthermore, while analyzing disparities in amputation rates, potential risk factors, such as age, sex, race, severity of illness, and risk of mortality, were considered.

We utilized data from the New York Statewide Planning and Research Cooperative System (SPARCS) for the years 2019 (pre-COVID) and 2021 (post-COVID), focusing on individuals primarily diagnosed with diabetes mellitus. To evaluate variations in

amputation rates, both direct and indirect standardization methods were employed. To identify potential individual risk factors, such as age, sex, race, ethnicity, and hospital-level characteristics, multivariate logistic regression models and Generalized Estimating Equations (GEE) models were used. Candidate models were compared using the Likelihood Ratio Test, Akaike Information Criterion (AIC), accuracy, and the Youden index to select the most accurate model. Lastly, Receiver Operating Characteristic (ROC) curves were used to assess the models' sensitivity and specificity.

In 2019, a total of 44,377 patients were diagnosed with diabetes, and 5,004 of them underwent LEA. In 2021, there were 40,573 diabetic patients, with 4,432 undergoing LEA. We expect that hospital-specific amputation rates will generally be higher after the pandemic, after controlling for patient- and hospital-level covariates. Due to the increasing hurdles to hospital admission for persons with diabetes during the COVID-19 pandemic, we anticipate an increase in the average severity of admitted patients and, as a result, higher amputation rates. Moreover, we expect the multivariate regression models to show that older age, male sex, and non-white race are associated with greater amputation rates.

The study's purpose was to determine how COVID-19 affected the incidence of LEA in diabetic patients in New York State hospitals. By evaluating pre- and post-COVID standardized amputation rates using multiple regression models, the results revealed a substantial rise in amputations following the pandemic outbreak. These findings suggest that COVID-19 may have had a deleterious impact on diabetic patient care. They underscore the importance of monitoring diabetic patient outcomes during crises and maintaining proper treatment measures.

Keywords

diabetes, lower-extremity amputation, COVID-19, hospital disparities, county-level analysis

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Chapter 1 - Introduction

Diabetes mellitus is a global public health concern, with the prevalence of this chronic disease increasing at an alarming rate worldwide. According to the International Diabetes Federation, the total number of people with diabetes in 2021 was 537 million, and this number is expected to increase to 783 million by 2045 [15]. Diabetic foot ulcer (DFU) is one of the most common complications of diabetes mellitus and represents the primary risk factor for lower extremity amputations (LEAs) [1].

LEAs can be classified into major (amputation at the ankle, through the knee, or up to the hip) and minor (amputation of the foot or toes). Diabetes-related LEAs account for three-quarters of all LEAs worldwide [11], resulting in reduced quality of life, increased medical costs, and a high risk of mortality following surgery, both in the short and long term. A national study in New Zealand, which focused on individuals with diabetes who had undergone major amputations, found that over 11% of patients had died within 30 days of surgery, while nearly 18% had died within 90 days [7]. A systematic review found very high 5-year mortality rates among patients with major amputations, ranging from 52% to 80% [8]. These findings underscore the importance of preventing the progression of diabetic complications through timely and effective management before amputation becomes necessary.

It is widely recognized that quality healthcare services can delay diabetes-related complications and play a vital role in improving a patient's quality of life [9]. However, amputation rates as a healthcare indicator must be treated with caution, as amputations, in opposition to limb salvage, can be either beneficial or harmful, depending on the patient's symptoms [20]. Problematic high amputation rates were already publicly recognized by the WHO in 1990 within the St. Vincent Declaration [21], which aimed to improve quality of life by reducing diabetes-related complications and halving the number of subsequent amputations [18]. Thus, LEA rates are an important indicator of the success of preventive care but do not directly reflect acute care performance.

Given the importance of regular monitoring, the COVID-19 pandemic represented a severe hurdle to the delivery of quality healthcare for patients with diabetes and DFU. Lockdowns and public health measures implemented during the pandemic restricted access to routine diabetes care, limiting new diagnoses and affecting self-care practices and access to medications [10]. A decrease in surveillance likely influenced amputation rates among patients with diabetes during the pandemic. According to a survey of healthcare professionals from 47 countries, diabetes was the most impacted chronic condition during the COVID-19 pandemic due to disruptions in care [11].

Several studies investigating the impact of COVID-19 on amputation incidence, comparing data from before and during the pandemic, found a significant increase in the ratio of major to minor lower extremity amputations. In Italy, Caruso et al. [13] found that patients admitted to the hospital for DFU in 2020 were more than three times as likely to undergo an amputation compared to patients in 2019. These results are supported by a single-center study from the Netherlands [6], which found that in 2020 patients exhibited more severe cases of peripheral arterial disease, and the number of amputations performed was threefold greater than in 2019. Similar findings for major LEA were reported in Germany [16]. In contrast, other studies found no significant difference in diabetes-related amputation rates [14].

Objectives

The aim of this paper is to assess the indirect impact of COVID-19 on diabetes-related LEAs using a large sample size, examining the variation in amputation rates between 2019 and 2021 among hospitals in the State of New York. Many of the available publications are single-institution studies with small sample sizes that focus only on the initial surge of the pandemic in 2020. In contrast, this study compares data from 2019 and 2021, after the first wave of COVID-19, to provide a broader contribution to existing knowledge by exploring the progression of the phenomenon over time.

Chapter 2 - Materials and Methods

To study the variation in amputation rates among diabetic patients before and after the surge of the COVID-19 pandemic, we utilized two datasets of hospital inpatient discharges from the New York Statewide Planning and Research Cooperative System (SPARCS) for the years 2019 (pre-COVID) and 2021 (post-COVID). These datasets included both demographic and clinical information for each patient discharged from hospitals in New York State. We selected only patients with a primary diagnosis of diabetes mellitus, with or without complications, resulting in sample sizes of $N = 44,377$ in 2019 and $N = 40,573$ in 2021. Unless otherwise specified, the reference population of this study consists of diabetic patients.

To determine how many diabetic patients underwent LEA, we selected individuals treated with one of the following procedures:

- Above-knee and other proximal lower extremity amputation,
- Below-knee amputation,
- Toe and mid-foot amputation.

We computed an indicator for the amputation rate to measure the impact of COVID-19 on diabetes mellitus treatments. The rate was calculated as the ratio of diabetic patients who underwent LEA to the total number of diabetic patients admitted to the hospital:

$$\text{Amputation Rate} = \frac{\text{Number of diabetic patients who underwent LEA}}{\text{Number of diabetic patients}}$$

For each patient, we recorded the hospital from which they were discharged and the county in which the hospital was located. Additionally, we included information on the patients' age group, sex, and race. The literature indicates that males are statistically 1.3 times more likely than females to develop diabetes complications resulting in LEA [17]. Both sex and age have been reported as significant risk factors for amputations due to diabetes complications in epidemiologic studies [19]. Furthermore, race has been

shown to affect the probability of LEA in diabetic patients. Following the recommendations of a systematic review [17] of 21 studies on LEA risk factors, we ignored the distinction between type 1 and type 2 diabetes. Instead, we included variables for severity of illness and risk of mortality, both ranging from minor to extreme. Patients with missing information on any of these predictors were excluded from the analysis.

Exploratory Analysis

We first reported descriptive statistics to examine the demographic characteristics of the patients, their clinical status, and the variation in crude amputation rates among diabetic patients in hospitals and counties of New York State.

To assess variability in amputation rates among hospitals and the changes following the COVID-19 pandemic, we applied both direct and indirect standardization. These methods allowed us to account for differences in population composition, specifically age and sex, across hospitals and counties.

We calculated directly standardized amputation rates (DAR) by dividing the expected number of amputations in a standard population by the size of the standard population:

$$DAR = \frac{\sum_{ij} N_{ij} \frac{a_{ij}}{n_{ij}}}{N}$$

where $\frac{N_{ij}}{N}$ represents the structure of the standard population, a_{ij} is the number of amputations, and n_{ij} is the number of diabetic patients in each hospital or county. By removing the confounding effects of age and sex, these rates facilitated comparisons across hospitals and counties.

We calculated indirectly standardized amputation rates (IAR) as the ratio of observed amputations to those expected if the population had the same age- and sex-specific rates as a standard population:

$$IAR = \frac{\sum_{ij} a_{ij}}{\sum_{ij} n_i \frac{A_i}{N_i}}$$

Here, the numerator represents the observed amputations, and the denominator represents the expected amputations. Unlike direct standardization, IAR does not allow direct comparisons between hospitals or counties but can be compared to the overall average amputation rate in the diabetic population.

We used these standardized rates to identify hospitals and counties with the greatest differences in amputation rates between 2019 and 2021.

Risk Factor Analysis

Following the exploratory analyses, we constructed multivariate regression models to identify potential risk factors for amputations. As the dependent variable (LEA) is dichotomous, we performed a binomial regression model from the family of Generalized Linear Models (GLM) with a logit link function. We compared these results with Generalized Estimating Equation (GEE) models, which account for potential correlations within data, particularly within hospitals. For GEE models, we specified an exchangeable correlation matrix, assuming equal correlation among responses within a cluster.

For both GLM and GEE models, we fitted a full model including demographic (age, sex, race) and clinical variables (severity of illness, risk of mortality), and a reduced model including only demographic variables.

Model Evaluation

To compare the models, we used several approaches:

- Akaike Information Criterion (AIC) to evaluate model complexity and fit,
- Likelihood Ratio Test (for GLM models),
- Receiver Operating Characteristic (ROC) curves and the Youden index to assess sensitivity and specificity.

The Area Under the ROC Curve (AUC) measures the model's discrimination ability, where higher AUC values indicate better performance. For GEE models, likelihood-based methods such as the Likelihood Ratio Test were excluded since GEE does not specify a likelihood function.

Chapter 3 - Results

Descriptive Statistics

In 2019, there were 44,377 patients hospitalized with a primary diagnosis of diabetes mellitus, while in 2021 the number decreased to 40,573. The distributions of the main demographic and clinical characteristics of these patients are reported in Table 1. Among diabetic patients in 2019, 5,004 (11.28%) underwent lower extremity amputation (LEA), while in 2021, 4,432 (10.92%) diabetic patients underwent the procedure.

Table 1. Demographic and clinical characteristics of patients.

Variable	Status	N (frequency %) 2019	N (frequency %) 2021
LEA	No	39,373 (88.72%)	36,141 (89.08%)
	Yes	5,004 (11.28%)	4,432 (10.92%)
Age group	0-17	1,564 (3.52%)	1,662 (4.10%)
	18-29	3,401 (7.66%)	3,076 (7.58%)
	30-49	8,566 (19.30%)	7,788 (19.20%)
	50-69	18,646 (42.02%)	17,000 (41.90%)
	70 or older	12,200 (27.49%)	11,047 (27.23%)

Sex	Female	18,616 (41.95%)	16,737 (41.25%)
	Male	25,761 (58.05%)	23,836 (58.75%)
Race	White	19,431 (43.79%)	17,484 (43.09%)
	Black/African American	13,310 (29.99%)	12,291 (30.29%)
	Multi-racial	537 (1.21%)	415 (1.02%)
	Other Race	11,099 (25.01%)	10,383 (25.59%)
Severity of diabetes	Minor	4,307 (9.71%)	5,439 (13.41%)
	Moderate	16,701 (37.63%)	19,506 (48.08%)
	Major	19,765 (44.54%)	13,171 (32.46%)
	Extreme	3,604 (8.12%)	2,457 (6.606%)
Risk of mortality	Minor	12,764 (28.76%)	10,382 (25.59%)
	Moderate	18,096 (40.78%)	15,403 (37.96%)
	Major	19,765 (44.54%)	12,801 (31.55%)
	Extreme	3,604 (8.12%)	1,987 (4.90%)

The analysis of individual demographic characteristics revealed that males were more likely to undergo LEA than females, a finding consistent with previous studies, such as Lin et al. [17]. Regarding age, the frequency of LEA was low in patients under the age of 50. The highest number of amputations was observed in the 50–69 age group, while patients aged 70 or older were less likely to undergo LEA. This could be due to the increased risks associated with performing the procedure on older patients. Due to the small number of patients younger than 30 years old, the two youngest age groups (0–17 and 18–29) were combined in subsequent analyses to ensure more meaningful results.

When considering race, White patients were the most frequently amputated in both 2019 and 2021, followed by Black or African American patients. Patients of other races, including multiracial individuals, underwent LEA less frequently in comparison.

The severity of diabetes was strongly associated with the likelihood of undergoing LEA. Patients diagnosed with diabetes of major or extreme severity were significantly more likely to require amputation. However, the frequency of LEA decreased for patients with extreme severity in 2021 compared to 2019. The risk of mortality exhibited a similar trend. Patients with a minor risk of mortality were the least frequently amputated, while those with a major risk of mortality were the most frequently subjected to the procedure. Interestingly, patients with an extreme risk of mortality were less likely to undergo LEA, which could be attributed to the high procedural risks associated with their condition.

This descriptive analysis demonstrates clear patterns in the demographic and clinical characteristics of diabetic patients undergoing LEA. It highlights the significant roles of age, sex, race, and clinical severity in influencing the likelihood of amputation. These results set the foundation for further bivariate and multivariate analyses, which are detailed in subsequent sections.

Standardization of Amputation Rates

To enable meaningful comparisons of amputation rates across hospitals and counties, we computed standardized rates using both direct and indirect standardization methods. These rates were adjusted for age group and gender to account for demographic differences among the populations served by various hospitals and counties.

Direct Standardization

The direct standardization approach was used to compare amputation rates between different hospitals and counties and to analyze the changes in these rates before (2019) and after (2021) the outbreak of the COVID-19 pandemic.

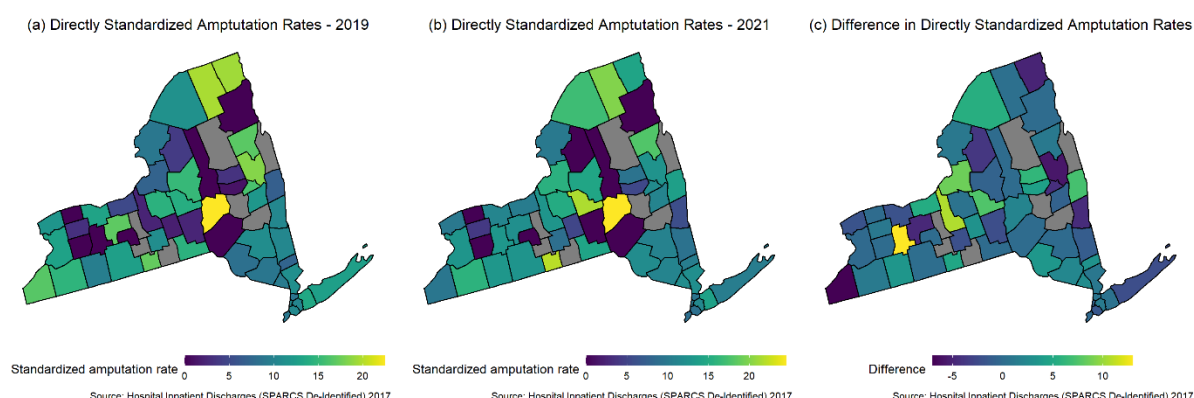
At the hospital level, 24 hospitals reported no LEAs in either 2019 or 2021, resulting in a null amputation rate. Among the hospitals with reported LEAs, Plainview Hospital had the highest standardized amputation rate in both years. In 2019, it was followed by the Health Alliance Hospital Mary's Avenue Campus and the Mary Imogene Bassett Hospital, with the latter maintaining its second-place ranking in 2021.

When comparing changes between 2019 and 2021, the amputation rate decreased in 102 hospitals but increased in 78 hospitals. Notable decreases were observed at UPMC Chautauqua at WCA (a decrease of 12.83), Glen Cove Hospital (a decrease of 9.96), and F. F. Thompson Hospital (a decrease of 9.11). Conversely, hospitals like Montefiore Mount Vernon Hospital and Auburn Hospital experienced increases in their standardized amputation rates.

At the county level, Otsego and Franklin counties had the highest standardized amputation rates in both 2019 and 2021. In 2019, these were followed by Clinton and Saratoga counties, while Chemung and Madison counties emerged as having the next highest rates in 2021. Figure 1 illustrates the magnitude of directly standardized rates at the county level, with panel (a) representing 2019, panel (b) 2021, and panel (c) showing the differences between the two years.

In terms of changes between 2019 and 2021, amputation rates decreased in 23 counties but increased in 26. Chautauqua and Saratoga counties saw the largest decreases, with reductions of 6.99 and 6.38, respectively, followed by Clinton and Ontario counties. On the other hand, Livingston and Cayuga counties exhibited the largest increases, with rates rising by 12.88 and 10.58, respectively, as shown in Figure 1 panel (c).

Figure 1. Directly standardized amputation rates at a county level (a) in 2019, (b) in 2021, and (c) their difference.



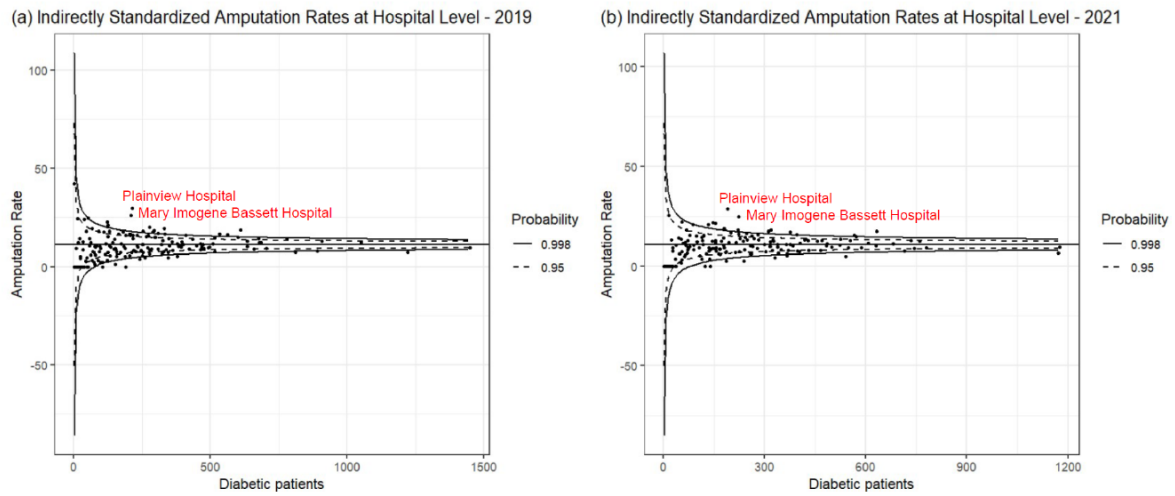
Indirect Standardization

Using the indirect standardization approach, we compared the amputation rates of each hospital to the crude amputation rate across the entire diabetic population, which was 11.28% in 2019 and 10.92% in 2021. This method allowed us to identify hospitals where the standardized amputation rate was significantly higher than the average.

In both 2019 and 2021, Plainview Hospital and Mary Imogene Bassett Hospital had the highest excessive amputation rates. In 2019, these hospitals exceeded the average amputation rate by 18.43 and 14.74 percentage points, respectively. In 2021, their excessive rates were slightly lower, at 17.84 and 14.00 percentage points, respectively.

Figure 2 presents the results in the form of funnel plots, which compare the indirectly standardized amputation rates to the number of diabetic patients for each hospital. These plots serve as a graphical tool to assess hospital performance in managing diabetic patients. Panel (a) represents 2019, while panel (b) illustrates 2021.

Figure 2. Funnel plots of indirectly standardized amputation rates at the hospital level (a) in 2019 and (b) in 2021.



GLMs and GEE Models

Based on the exploratory analysis, we constructed a full model that included age, sex, race, severity of diabetes, risk of mortality, and year as predictors. A reduced model excluded the severity of diabetes and risk of mortality variables. These models were designed to quantify risk factors for lower extremity amputations (LEA) among diabetic patients. We employed both Generalized Linear Models (GLMs) and Generalized Estimating Equation (GEE) models, using identical predictors, to assess the potential influence of correlations among patients treated within the same hospital.

The full model was defined as:

$$LEA \sim \beta_0 + \beta_1 Age + \beta_2 Sex + \beta_3 Race + \beta_4 Severity + \beta_5 Risk + \beta_6 Year$$

All predictors, except for the "multi-racial" race category, were found to have a statistically significant effect on the likelihood of undergoing LEA. The magnitude and direction of the effects were consistent between the GLM and GEE models, as summarized in Table 2.

Table 2. Odds Ratios of the full GLM and GEE model.

Variable	Status	GLM		GEE	
		OR (95% CI)	p-value	OR (95% CI)	p-value
Year	2021	1.30 (1.24, 1.37)	0.00	1.30 (1.24, 1.36)	0.00
Age group	30-49	16.30 (12.15, 22.61)	0.00	16.23 (11.89, 22.14)	0.00
	50-69	34.80 (25.9, 48.02)	0.00	34.59 (25.42, 47.07)	0.00
	70 or older	34.85 (25.96, 48.18)	0.00	34.64 (25.41, 47.21)	0.00
Sex	Male	2.02 (1.92, 2.13)	0.00	2.02 (1.93, 2.13)	0.00
Race	White	1.40 (1.33, 1.48)	0.00	1.40 (1.32, 1.48)	0.00
	Multi-racial	1.07 (0.83, 1.35)	0.59	1.06 (0.83, 1.36)	0.62
	Other Race	1.18 (1.11, 1.26)	0.00	1.1 (1.11, 1.26)	0.00
Severity of diabetes	Moderate	13.04 (10.39, 16.55)	0.00	12.98 (10.29, 16.390)	0.00
	Major	45.67 (36.32, 58.38)	0.00	45.48 (35.95, 57.556)	0.00
	Extreme	138.05 (108.20, 178.74)	0.00	137.42 (107.01, 176.49)	0.00

Risk of mortality	Moderate	0.48 (0.45, 0.51)	0.00	0.48 (0.45, 0.51)	0.00
	Major	0.27 (0.25, 0.29)	0.00	0.27 (0.24, 0.29)	0.00
	Extreme	0.10 (0.09, 0.11)	0.00	0.10 (0.09, 0.11)	0.00

The risk of amputation increased significantly with age. Patients aged 50 or older had a 34-fold higher risk of undergoing amputation compared to those under 30, while patients aged 30 to 49 had a 16-fold increase in risk. Males were twice as likely to undergo LEA as females, with an odds ratio (OR) of 2.02. Race also influenced the risk, with White patients showing a 40% higher likelihood of amputation compared to African American patients. Individuals of other races demonstrated an 18% increased risk, while the multi-racial category did not show a significant difference in risk compared to African Americans.

Severity of diabetes was a major determinant of amputation risk. The likelihood of LEA increased dramatically with the severity of the condition, as patients with extreme severity were over 138 times more likely to undergo the procedure compared to those with minor severity. Conversely, the risk of mortality at admission exhibited an inverse relationship with the likelihood of amputation. Patients with an extreme risk of mortality were 90% less likely to undergo LEA compared to those with minor risk, possibly due to the heightened surgical risks associated with critically ill patients. Additionally, there was a 30% increase in the risk of amputation in 2021 compared to 2019, underscoring the potential impact of the COVID-19 pandemic on diabetes care.

The GEE model, incorporating an exchangeable correlation matrix, reported an alpha value of 0.011, indicating weak correlations within the same hospital. This weak association justified the use of GLMs for subsequent analyses without significant loss of explanatory power.

Model Reduction and Validation

To assess the relative performance of the full and reduced models, we compared their predictive capabilities. The reduced model, excluding severity and risk of mortality, was defined as:

$$LEA \sim \beta_0 + \beta_1 Age + \beta_2 Sex + \beta_3 Race + \beta_4 Year$$

The performance of the two models was evaluated using the Akaike Information Criterion (AIC) and the likelihood ratio test. For the full model, the log-likelihood was

$$-2LL_{full} = 50886.65$$

while the reduced model had a log-likelihood of

$$-2LL_{reduced} = 55444.3$$

The likelihood ratio test yielded a difference of 4557.73, with a p-value < 0.05 , indicating that the full model provided a significantly better fit. These results were corroborated by the AIC values, where the full model outperformed the reduced model with values of 50,917 and 55,462, respectively.

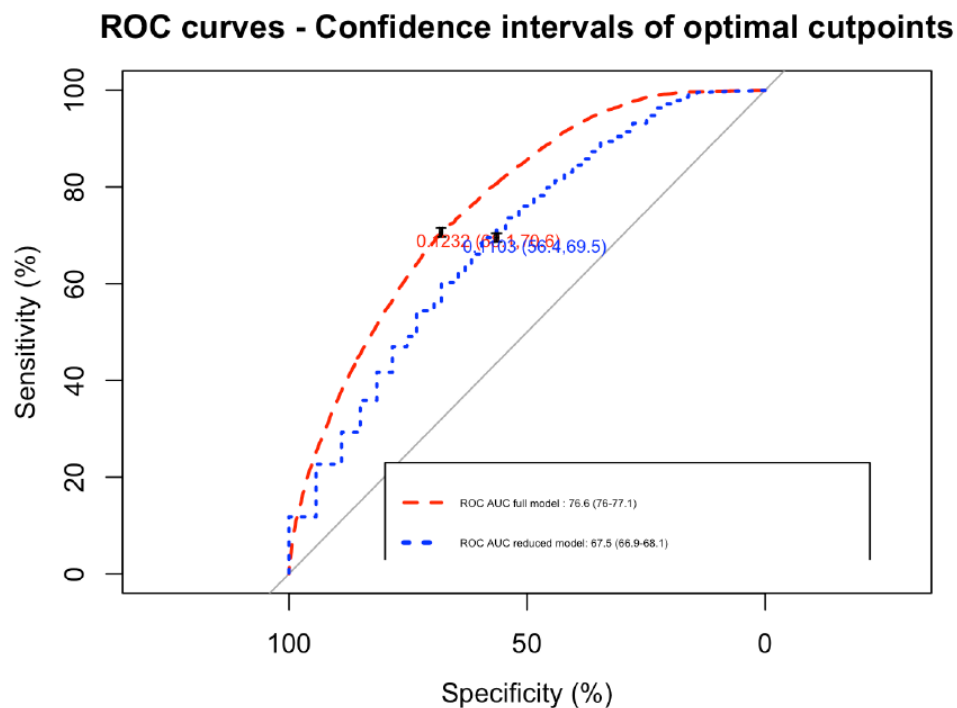
The predictive accuracy of the models was further assessed using ROC curves, represented in Figure 3. The full model achieved an Area Under the Curve (AUC) of 0.765, representing acceptable discrimination of the data, while the reduced model yielded a lower AUC of 0.675, indicating poorer performance. To refine the evaluation, we computed optimal cut-points using the Youden Index, which balances sensitivity and specificity. As reported in Table 3, the optimal cut-point for the full model was 0.123, yielding a sensitivity of 70.6% and specificity of 68.1%. In contrast, the reduced model's optimal cut-point of 0.110 resulted in a sensitivity of 69.5% but a significantly lower specificity of 56.4%.

Table 3. Optimal cut-points with regard to the Youden index and corresponding sensitivity and specificity values for the full and reduced model.

Model	Optimal cut-point	Youden	Accuracy	Sensitivity	Specificity
Full	0.123	0.387	0.684	0.706	0.681
Reduced	0.110	0.259	0.579	0.695	0.564

Given the imbalanced nature of the data, where the number of non-amputated patients was approximately nine times greater than amputated patients, a naive model predicting no amputations for all patients would achieve an accuracy of 88.72%. However, this approach would fail to provide meaningful insights into the determinants of LEA. By incorporating the Youden Index, we accounted for the imbalance and demonstrated that the full model provided superior discrimination, particularly in identifying true negatives.

Figure 3. Roc curves, optimal cutpoints and confidence intervals obtained through Youden Index.



The analysis underscored the need to evaluate both sensitivity and specificity, particularly in imbalanced datasets. The superior performance of the full model highlighted the importance of including clinical variables, such as severity of diabetes and risk of mortality, to accurately predict LEA outcomes.

Chapter 4 - Conclusions

This study revealed a significant increase in the risk of receiving a lower extremity amputation (LEA) among diabetic patients between 2019 and 2021, regardless of the model used (GLM or GEE). All predictors included in the models were significantly associated with the risk of LEA, with the robustness of the results further supported by the large sample size of the reference population. These findings extend those of previous studies [6, 13, 16], which primarily focused on the initial year of the pandemic. The observed rise in amputation rates during the early stages of the pandemic was largely attributed to reduced access to preventive healthcare services. Notably, our results demonstrate that even in 2021, when pandemic-related restrictions were considerably eased, the risk of undergoing an LEA increased by 30% compared to the pre-pandemic year of 2019.

This underscores the critical importance of consistent and uninterrupted healthcare services for preventing the progression of diabetic complications. The increase in amputation risk highlights how disruptions in care, even if temporary, can lead to long-term negative health outcomes for patients with chronic conditions such as diabetes.

While the focus of this study was on comparing pre- and post-COVID-19 periods, our findings also suggest that structural differences between hospitals in New York State have a relatively minor influence on amputation rates. Although a few outliers were identified, as indicated by the funnel plot analysis, the overall impact of hospital choice on LEA risk was negligible. The small alpha value from the full GEE model further supports this conclusion, indicating weak within-hospital correlation.

These results suggest that patient-specific factors, rather than hospital characteristics, play a more decisive role in determining amputation outcomes.

Limitations and Future Work

The retrospective nature of this study represents a key limitation, as it restricts the ability to make causal inferences and allows only correlation-based conclusions. Additionally, the absence of detailed data on relevant biometric variables and patient comorbidities limited the accuracy of the predictive models. Incorporating more granular patient information, such as HbA1c levels, duration of diabetes, or coexisting conditions like peripheral arterial disease, could improve the stratification of risks and deepen the understanding of the factors influencing LEA outcomes.

Another limitation is the inconsistent categorization of ethnicity between the datasets for 2019 and 2021, which precluded its inclusion as a predictor in the models. Ethnicity is known to play a significant role in healthcare outcomes, and future research should aim to resolve these inconsistencies to allow for a more comprehensive analysis.

Future studies could build on these findings by exploring longitudinal trends in amputation rates across additional years, extending beyond the pandemic period to capture long-term patterns. Moreover, prospective cohort studies could be conducted to establish causal relationships and assess the effectiveness of interventions designed to mitigate the risks of LEA among diabetic patients. These efforts could inform healthcare policies and promote strategies to ensure consistent care delivery, even in times of crisis.

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